

IMPROVING PROBABILISTIC DIFFUSION MODELS WITH OPTIMAL DIAGONAL COVARIANCE MATCHING

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Why covariance matters (and our goal)

- We denoise from Gaussian noise to data with diffusion, achieving strong fidelity across image, video, audio, and text.
- Accurate, state-dependent covariance accelerates sampling and improves likelihood.
- We target few-step generation where covariance choice dominates quality and NLL.
- Our goal: learn the optimal diagonal covariance efficiently.

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May 24, 2026

1 / 20

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May 24, 2026

2 / 20

Key result: We achieve a better quality–likelihood trade-off

- We dominate the Pareto frontier across steps on CIFAR10 (cosine schedule).

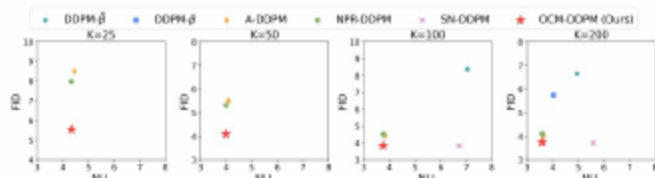


Figure: CIFAR10 (CS) 上, 不同采样步数的各法FID–NLL; 本方法权衡最优

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May 24, 2026

3 / 20

We reach $FID \approx 6$ in fewer steps

- We require the fewest steps on all datasets (after time-overhead normalization); 10 steps on ImageNet 256×256 (latent DiT).

Table: Table 4: The least number of timesteps ↓ required to achieve an FID around 6 (along with the corresponding FID).

METHOD	CIFAR10	CELEBA 64X64	LSUN BEDROOM	IMAGENET 256X256
DDPM	90 (6.12)	≥ 200	130 (6.06)	21 (5.89)
DDIM	30 (5.85)	≥ 100	BEST FID ≥ 6	11 (5.58)
IMPROVED DDPM	45 (5.96)	MISSING MODEL	90 (6.02)	22 (6.08)
ANALYTIC-DPM	25 (5.81)	55 (5.98)	100 (6.05)	MISSING MODEL
NPR-DPM	1.002×23 (5.76)	1.013×50 (6.04)	1.021×90 (6.01)	MISSING MODEL
SN-DPM	1.005×17 (5.81)	1.019×22 (5.96)	1.114×92 (6.02)	MISSING MODEL
OCM-DPM (OURS)	1.003×16 (5.83)	1.015×21 (5.94)	1.112×90 (6.04)	1.007×10 (5.33)

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May 24, 2026

4 / 20

We scale to latent ImageNet 256×256 (DiT)

- At 10 steps and $CFG \approx 2.0$, we achieve the best FID and higher Recall than Analytic-DDPM (A-DDPM), I-DDPM (Improved DDPM), DDPM- β / DDPM- $\tilde{\beta}$, and DDIM.

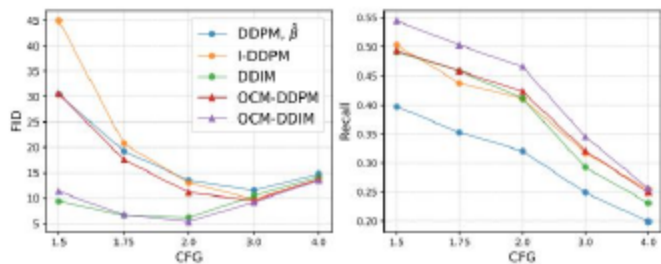


Figure: ImageNet 256×256上DiT训练结果：10步采样，可变CFG系数

The problem: We learn optimal, state-dependent diagonal covariance

- We aim to match the analytic optimal posterior covariance with minimal overhead.
- We face intractable Hessians at image scale; naive probes are too costly.
- We improve over fixed variance, biased diagonal fits, and heavy analytic estimators.

10-step CFG trade-off on ImageNet 256×256 (we outperform baselines)

- At 10 steps and $CFG \approx 2.0$, we outperform Analytic-DDPM (A-DDPM), I-DDPM (Improved DDPM), DDPM- β / DDPM- $\tilde{\beta}$, and DDIM on FID while improving Recall.

Table: Table 12: Results with varying CFG coefficients using 10 sampling steps on ImageNet 256x256.

	FID ↓					RECALL (%) ↑				
CFG =	1.5	1.75	2.0	3.0	4.0	1.5	1.75	2.0	3.0	4.0
DDPM, $\tilde{\beta}$	30.41	19.26	13.53	11.52	14.76	39.74	35.29	32.08	24.93	19.91
DDPM, β	210.28	182.78	158.16	89.86	58.16	16.98	22.28	25.13	23.07	19.48
I-DDPM	44.96	20.71	13.01	9.76	13.75	50.32	43.75	41.08	31.77	25.49
OCM-DDPM	30.55	17.57	11.12	9.52	13.74	49.39	45.87	42.33	32.13	24.96
DDIM	9.41	6.54	6.12	10.49	14.18	49.15	45.82	41.15	29.27	23.16
OCM-DDIM	11.30	6.67	5.33	9.20	13.49	54.50	50.32	46.58	34.53	25.67

We introduce OCM and our contributions

- We propose Optimal Covariance Matching (OCM) to regress optimal diagonal covariance from the score.
- We design an unbiased objective to learn the Hessian diagonal with one extra head/pass.
- We integrate into DDPM/DDIM and improve FID, Recall, NLL, and step count.

- We leverage an identity linking the score Hessian to the optimal posterior covariance.
- We train a diagonal-Hessian head $h_\phi(x_t, t)$ with an unbiased OCM loss.
- We plug the predicted diagonal into closed-form DDPM/DDIM mean/variance updates.

$$\Sigma^*(x_t) = \frac{\sigma_t^4 \nabla^2 \log q(x_t) + \sigma_t^2 I}{\alpha_t^2} \Rightarrow \text{diag}(\Sigma^*(x_t)) = \frac{\sigma_t^4 \text{diag}(\nabla^2 \log q(x_t)) + \sigma_t^2}{\alpha_t^2}.$$

- We interpret: σ_t (noise level) and α_t (data coefficient) scale variance; the Hessian diagonal of $\log q(x_t)$ sets state-dependent curvature.

- We note full Hessians are infeasible for images; exact diagonals need D passes.
- We avoid many Hutchinson/Rademacher probes that induce high variance and latency.
- We retain fast generation by amortizing curvature prediction.

- We train $h_\phi(x_t, t)$ via $\mathbb{E}[\|h_\phi(x_t, t) - v \odot (\nabla^2 \log q(x_t) v)\|^2]$, with v Rademacher.
- We upper-bound naive averaging and recover the true Hessian diagonal at optimum.
- We find $M=1$ probe per sample sufficient during training in practice.

- We reuse a pretrained UNet/DiT backbone and optionally freeze the score head.
- We add a small head predicting $\text{diag}(H)$ as $h_\phi(x_t, t)$.
- We incur negligible inference overhead ($\approx 0\text{--}11\%$ /step), offset by fewer steps.

- Flow (pseudo-diagram): Score $\rightarrow$ Hessian diag $h_\phi \rightarrow \Sigma(x_t; \phi) \rightarrow p_{\theta, \phi}$ update.
- We plug h_ϕ into analytic $\Sigma(x_t; \phi)$ to form state-dependent diagonal covariance.
- We use closed-form $p_{\theta, \phi}(x_{t'}|x_t)$ and $p_{\theta, \phi}(x_0|x_t)$ for DDPM/DDIM with Tweedie means.

- We upper-bound the grounded objective and attain the true $\text{diag}(H)$ at optimum.
- We explain SN via second-order Tweedie and avoid its quadratic error near $t \rightarrow 0$.
- We retain tractable likelihoods for a better NLL–quality trade-off.

- Datasets: CIFAR10, CelebA 64×64 , LSUN Bedroom, ImageNet 64×64 ; latent DiT on ImageNet 256×256 (LDM).
- Metrics: FID (quality), Recall (diversity), NLL (bits/dim).
- Baselines (naming consistent across slides/tables): DDPM- β ; DDPM- $\tilde{\beta}$; I-DDPM (Improved DDPM); DDIM; Analytic-DPM (A-DDPM); SN-DPM; NPR-DPM; Classifier-Free Guidance; DiT; LDM.

- We observe strong FID and notably better NLL at low steps, highlighting state-dependent covariance.

Table: Table 1: Comparison of different covariance choices on CIFAR10 (CS). We use a Rademacher estimator ($M = 100$) for diagonal covariance approximation in OCM-DDPM.

Method	5	10	15	20	5	10	15	20
DDPM, $\tilde{\beta}$	58.28	34.76	24.02	–	19.00	203.29	74.95	44.94
DDPM, β	–	–	–	–	7.33	6.51	6.06	5.77
OCM-DDPM	38.88	21.60	13.35	–	9.75	6.82	4.98	4.62

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- We learn optimal diagonal covariance via an unbiased objective, improving FID/Recall/NLL with negligible overhead.
- We integrate with DDPM/DDIM and scale to latent ImageNet (DiT), achieving target quality in fewer steps.
- Outlook: extend beyond diagonal (low-rank/block-diagonal), integrate with advanced ODE/SDE solvers and inverse problems, leverage better likelihoods for compression/science.

Questions & Discussion

- Thank you for your attention!
- We welcome your questions and feedback.

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Visual evidence (backup): we match target FID with fewer steps

- We match the target FID with fewer NFEs while maintaining diversity.



Figure: OCM-DPM最少步数下的训练数据与生成样本 (FID≈6, 见表4)